



Human Perception & Information Processing

Visualization of Information

Salomon Eisler

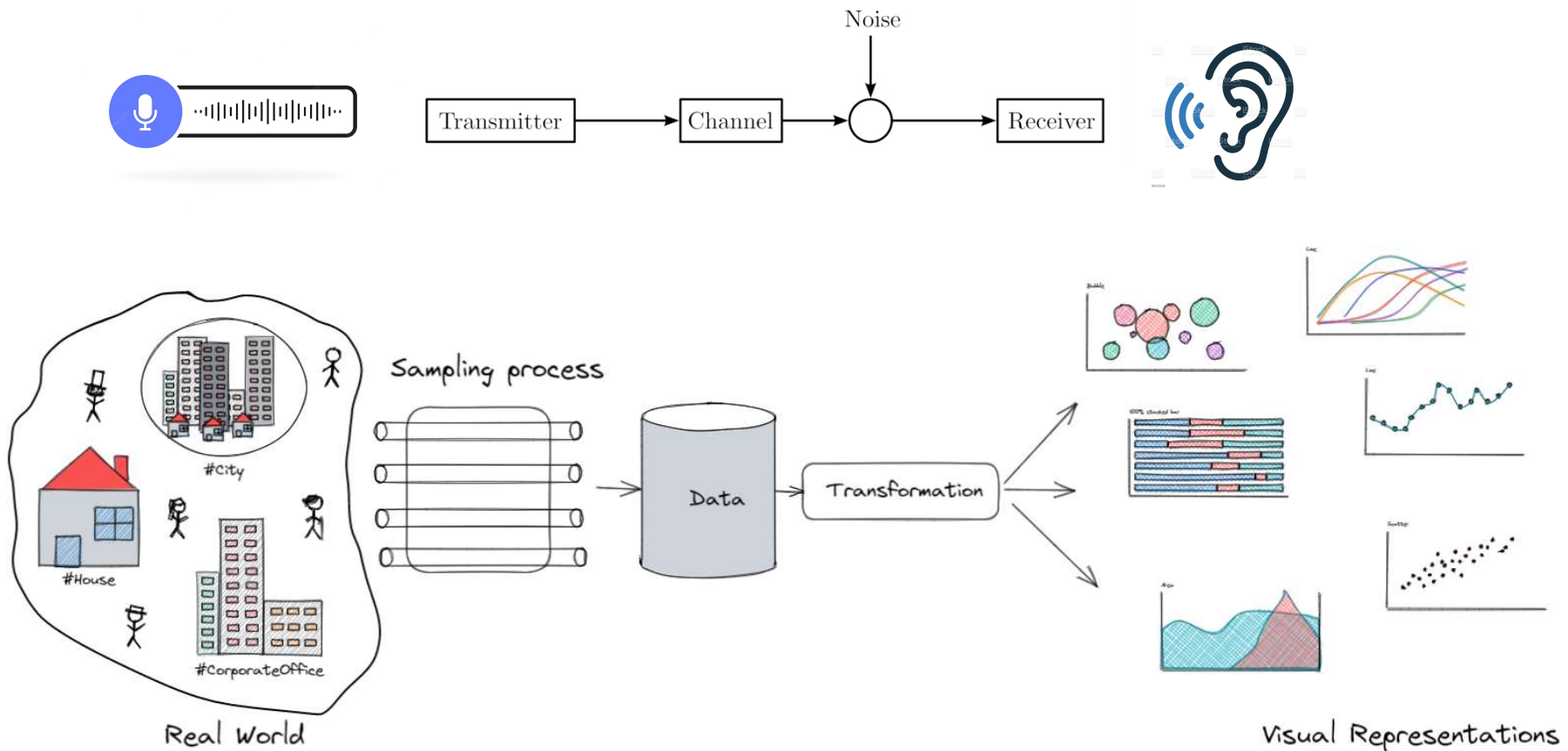
Dept. of Industrial Engineering

#	Content	Comment
1	High level overview of the steps of ML and discern when and why visualizations can be used to improve the process	Theory
2	Introduction: Why/What/How visualization pipe, semiology, variables, and taxonomy	Theory
3	Principles of human perception, principles of information processing and perception, saliency, etc.	Theory
4	Data sources and processes of data processing (pre-processing – data transformations)	Theory
5	Visualization methods for temporal data (time-series) and multi-dimensional data (multi variate)	Theory + Hands On
6	Interactive Visualizations with Shiny R (as an example) and other platforms (PowerBI) – how to use gen AI tools to generate visualizations	Theory + Hands On
7	In depth: Principles of interaction, methods of interaction in visualization	Theory + Hands On
8	Lie and truth in information presentation and Dashboards	Theory
9	Visualization of statistical information (e.g. variance, etc.) and visual analytics in machine learning	Theory + Hands On
10	Visualization of text and documents Visualization methods for trees, graphs, and networks	Theory + Hands On
11	Comparison and Evaluation of Visualizations + Story Telling	Theory + Hands On
12	Presentations	
13		

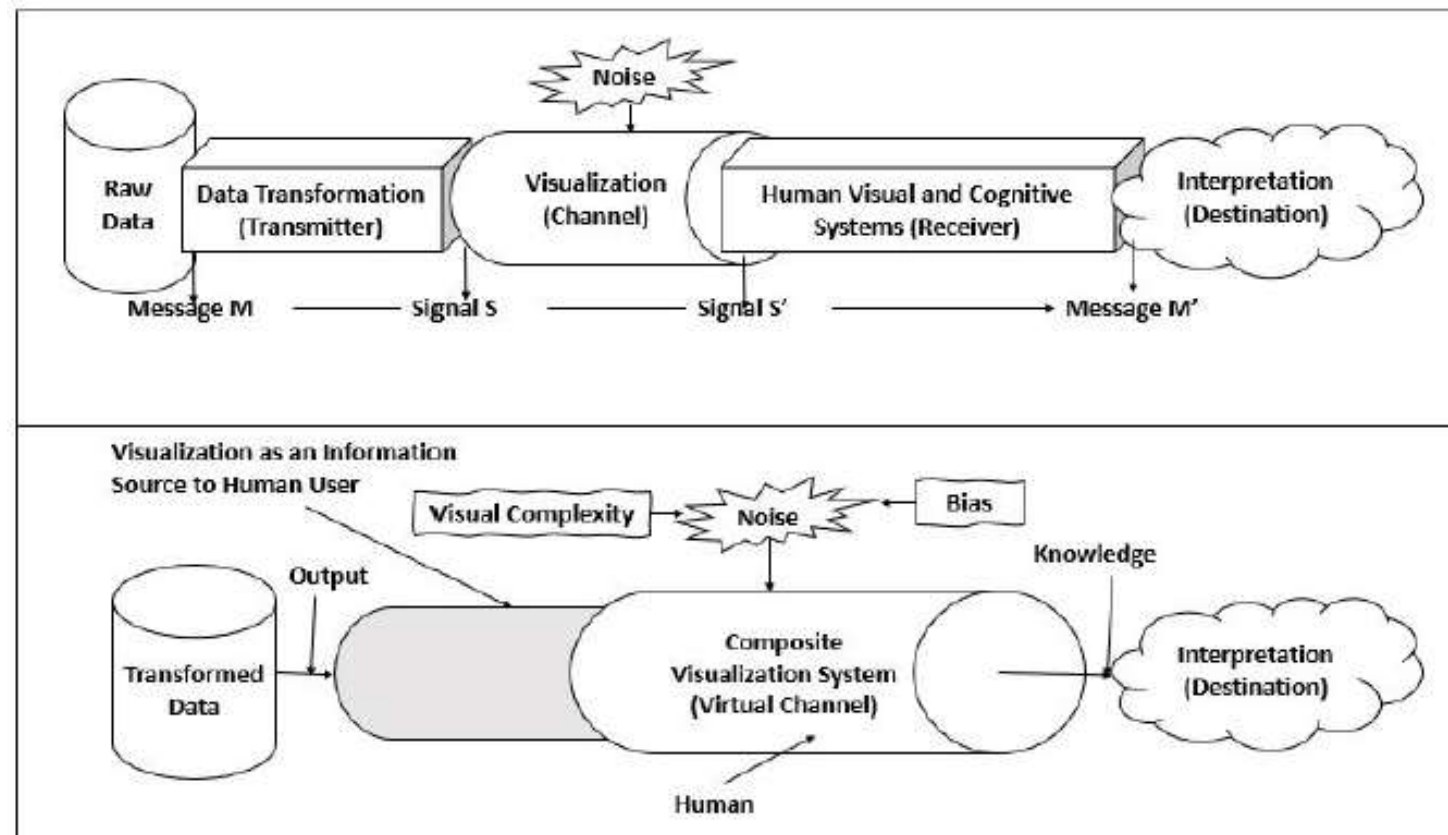
Agenda

1. Short review from last session – questions and answers
2. Why?
3. What Is Perception?
4. Perceptual Processing
5. Resource Model of Human Information Processing
6. Metrics
7. Cognition

Communication



Visualizations as communication channels



Semiology of Graphical Symbols

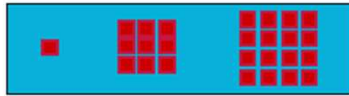
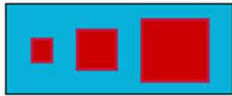
- To fully comprehend a sign system, one must first completely understand the primitive elements that define such a system.
- Bertin's graphical vocabulary, identifies some very specific graphical primitives that correspond to perceptual attributes, graphic qualities that are perceptually identifiable.
- Bertin notes that the plane "is so familiar that its properties seem self evident, but the most familiar things are often the most poorly understood";
- **Marks** Points, lines, and areas
- **Positional** Two planar dimensions
- **Retinal** Size, value, texture, color, orientation, and shape

The Eight Visual Variables - Bertin

The spatial arrangement of graphic primitives (marks) creates informative displays.



Position: changes in the x, y, (z) location



Size: change in length, area or repetition



Shape: infinite number of shapes



Value/brightness: changes from light to dark



Orientation: changes in alignment



Color: changes in hue at a given value



Texture: variation in pattern

Characteristics of visual variables

- Selective: is a change in this variable enough to allow us to select it from a group?
- Associative: is a change in this variable enough to allow us to perceive them as a group?
- Quantitative: is there a numerical reading obtainable from changes in this variable?
- Order: are changes in this variable perceived as ordered?
- Length: across how many changes in this variable are distinctions perceptible?

Shneiderman's Mantra

Visual Information Seeking Mantra:

- **Overview:** Gain an overview of the entire collection.
- **Zoom :** Zoom in on items of interest
- **Filter:** filter out uninteresting items.
- **Details-on-demand:** Select an item or group and get details when needed.
- **Relate:** View relationships among items.
- **History:** Keep a history of actions to support undo, replay, and progressive refinement.
- **Extract:** Allow extraction of sub-collections and of the query parameters

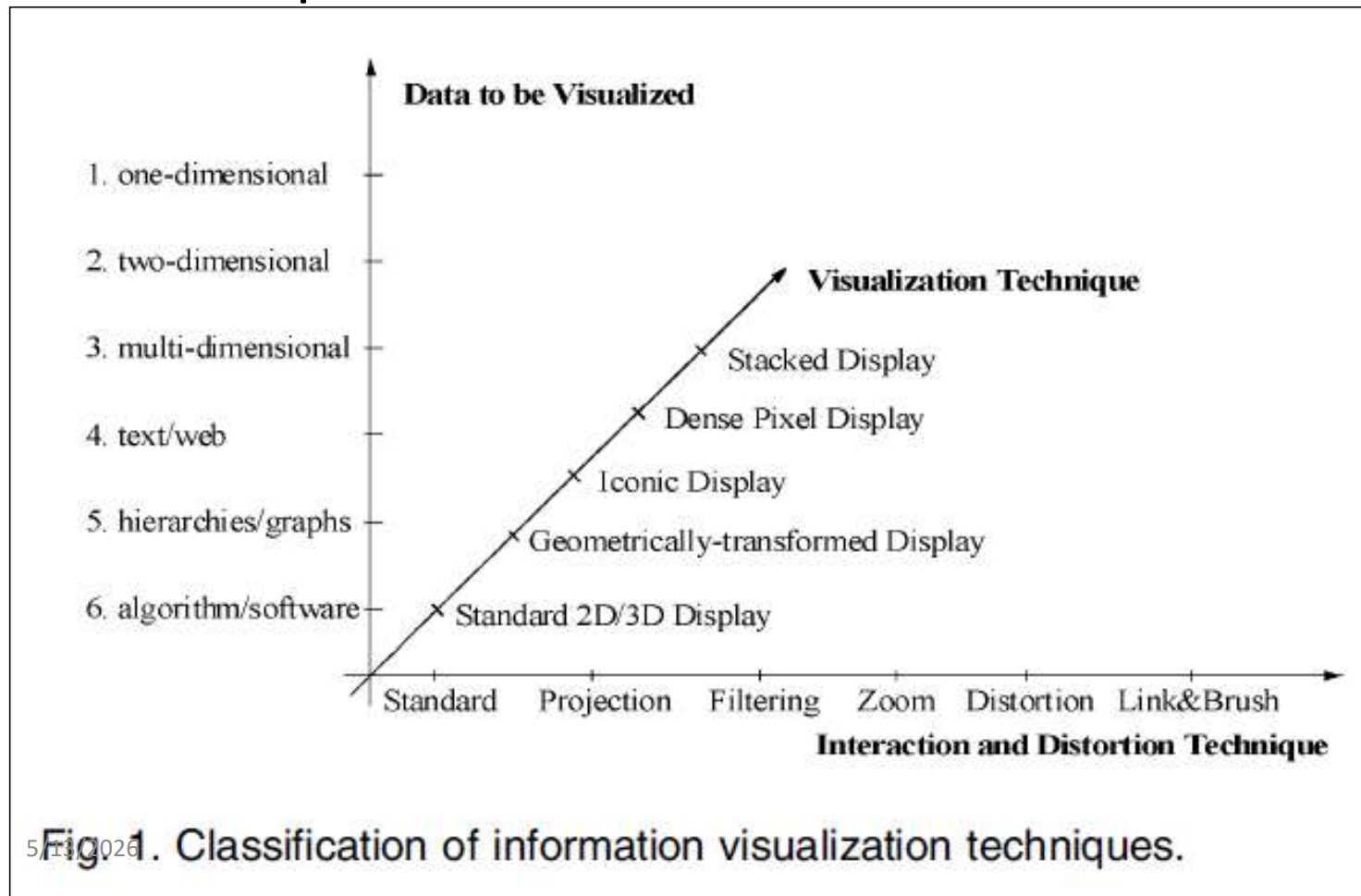
Shneiderman's 7 data types

#	Type	Short Description
1	1-dimensional	Linear data types include textual documents, program source code, and alphabetical lists of names which are all organized in a sequential manner. Also time series 😊
2	2-dimensional	Planar or map data include geographic maps, floorplans, or newspaper layouts. Each item in the collection covers some part of the total area and may be rectangular or not
3	3-dimensional	Real-world objects such as molecules, the human body, and buildings have items with volume and some potentially complex relationship with other items.
4	Temporal	Time lines are widely used and vital enough for medical records, project management, or historical presentations to create a data type that is separate from 1-dimensional data. The distinction in temporal data is that items have a start and finish time and that items may overlap.

Shneiderman's 7 data types

#	Type	Short Description
5	Multi-dimensional	most relational and statistical databases are conveniently manipulated as multidimensional data in which items with n attributes become points in a n -dimensional space.
6	Tree	Hierarchies or tree structures are collections of items with each item having a link to one parent item (except the root). Items and the links between parent and child can have multiple attributes.
7	Network	Sometimes relationships among items cannot be conveniently captured with a tree structure and it is useful to have items linked to an arbitrary number of other items.

Taxonomies – classifications of visualization techniques

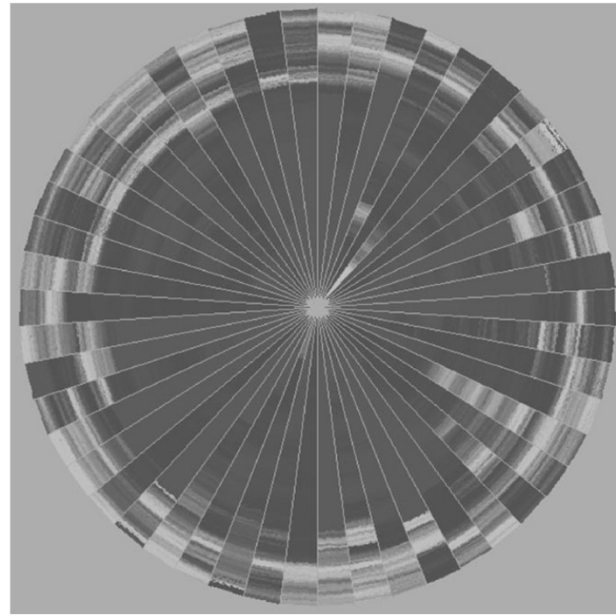
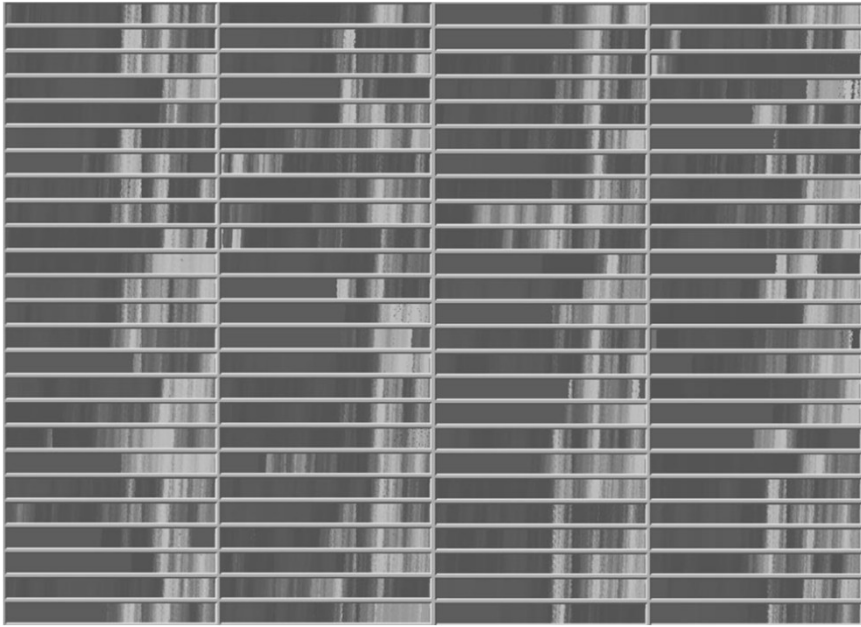


Keim (2002) Information
Visualization Classification

Information
Visualization and Visual
Data Mining
Daniel A. Keim, - 2002

Fig. 1. Classification of information visualization techniques.

Keim: Visualization techniques



The visualization shows 20 years (January 1974-April 1995) of daily prices of the 100 stocks contained in the Frankfurt Stock Index (FAZ).

Fig. 3. Dense pixel displays: recursive pattern technique

Fig. 4. Dense pixel displays: circle segments technique

The basic idea of **dense pixel techniques** is to map each dimension value to a colored pixel and group the pixels belonging to each dimension into adjacent areas.

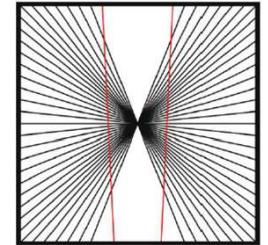
Papers to read

- The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations – Ben Schneiderman(1996)
- Information Visualization and Visual Data Mining - Daniel A. Keim (2002)
- Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods - WILLIAM S. CLEVELAND and ROBERT MCGILL*

Why perception?

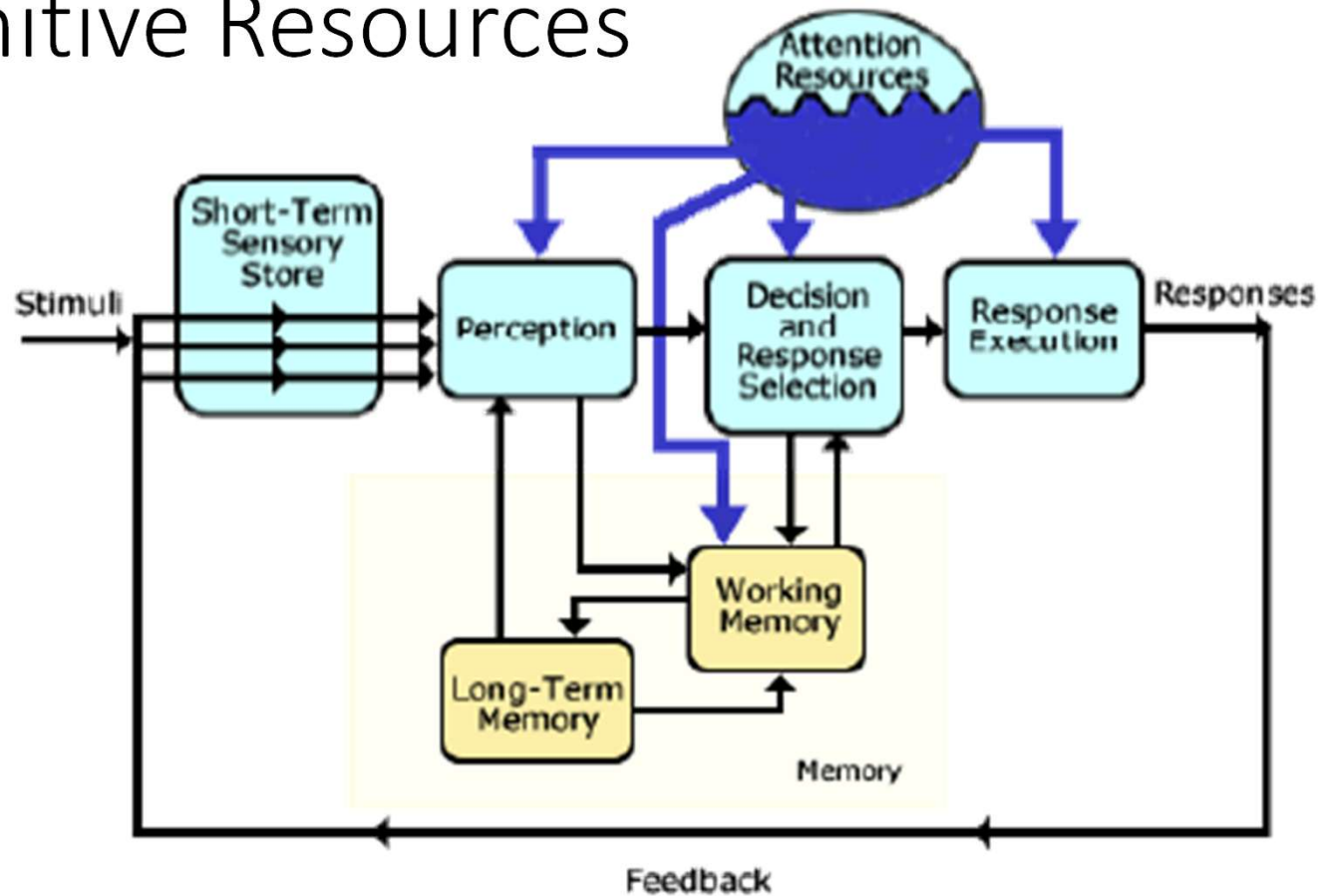
- Visualizations exist only when “seers/watcher” exist.
- Eyes and vision are parts of our perceptual systems.
- Visualizations mean nothing without perception.

What Is Perception?



- One definition: the process of recognizing (being aware of), organizing (gathering and storing), and interpreting (binding to knowledge) **sensory** information.
 - Interpret the world around us, forming a **mental representation** of the environment.
 - Visual representations of objects are often misinterpreted, either because they do not match our perceptual system, our mental models or **they were intended to be misinterpreted**.
- ➔ We need to understand sensory information and mental models.

Cognitive Resources



What is Psychophysics?

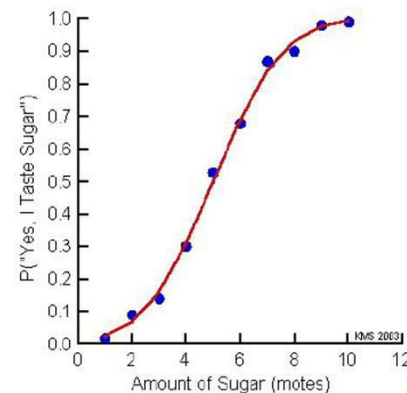
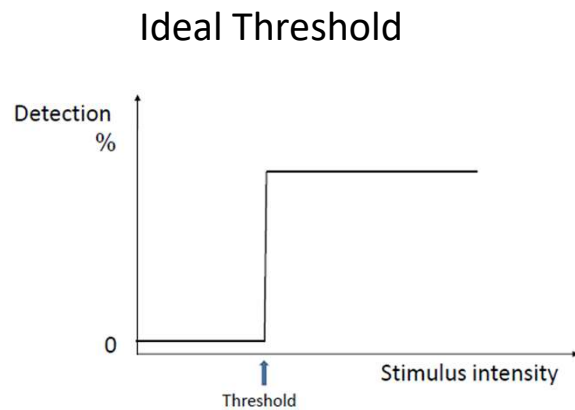
Psychophysics is the scientific study of the relationship between stimuli (specified in physical terms) and the sensations and perceptions evoked by these stimuli.
(narrow definition)

What is Psychophysics?

Psychophysics is commonly defined as the quantitative branch of the study of perception, examining the relations between observed stimuli and responses and the reasons for those relations.
(broader definition).

What is Psychophysics?

Since its inception, psychophysics has assumed that the human perceptual system is a measuring instrument yielding results (experiences, judgments, responses) that may be systematically analyzed.



Thresholds are defined as percentages. Typically, the threshold will be defined as the value for which the detection rate is 0.5. However, other thresholds may be used, depending on the purpose.

Absolute threshold: the smallest stimulus intensity that can be detected by an organism.

Absolute Thresholds (approximately)

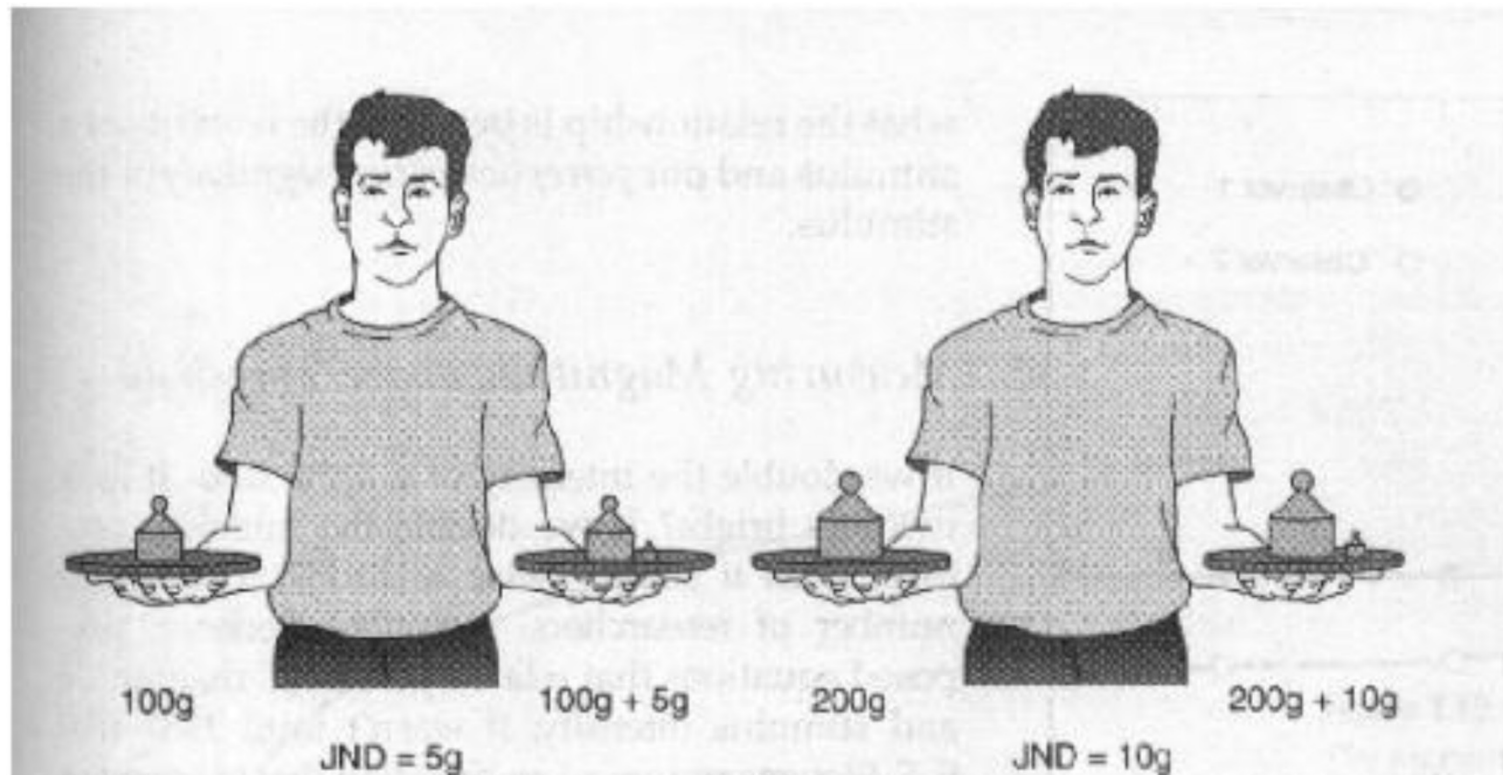
- Vision We can see 1 candle 30 miles away --(pretty low threshold!).
- Audition We can hear a watch tick 20 feet away.
- Taste We can taste 1 tsp. of sugar in 2 gallons of water.
- Smell We can smell 1 drop of perfume within a 3 room apartment.
- Touch We can feel the sensation of a bee wing dropped from 1 cm above your back.

Weber's Law

- Weber, Ernst Heinrich (1795-1878) Physician, anatomist and experimental psychologist.
- Defined Weber's law, specifying that the ratio of the just noticeable difference "j.n.d ." and the stimulus magnitude is a constant.



Just Noticeable Difference (jnd)



Weber Law

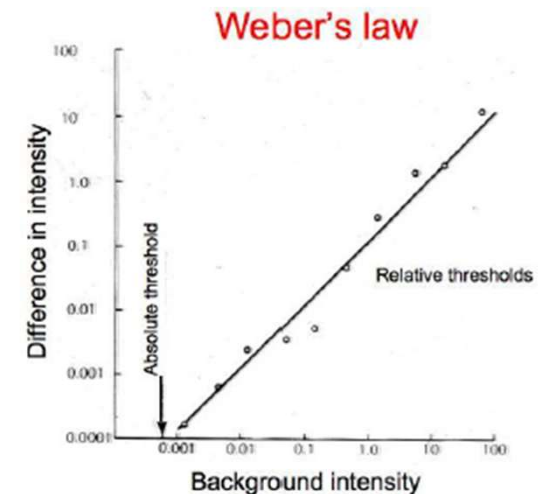
- Just Noticeable Difference (jnd): The minimal change in stimulus intensity required to exceed the differential threshold.
- The jnd is a constant fraction of the stimulus intensity

$$\frac{\Delta R}{R} = k$$

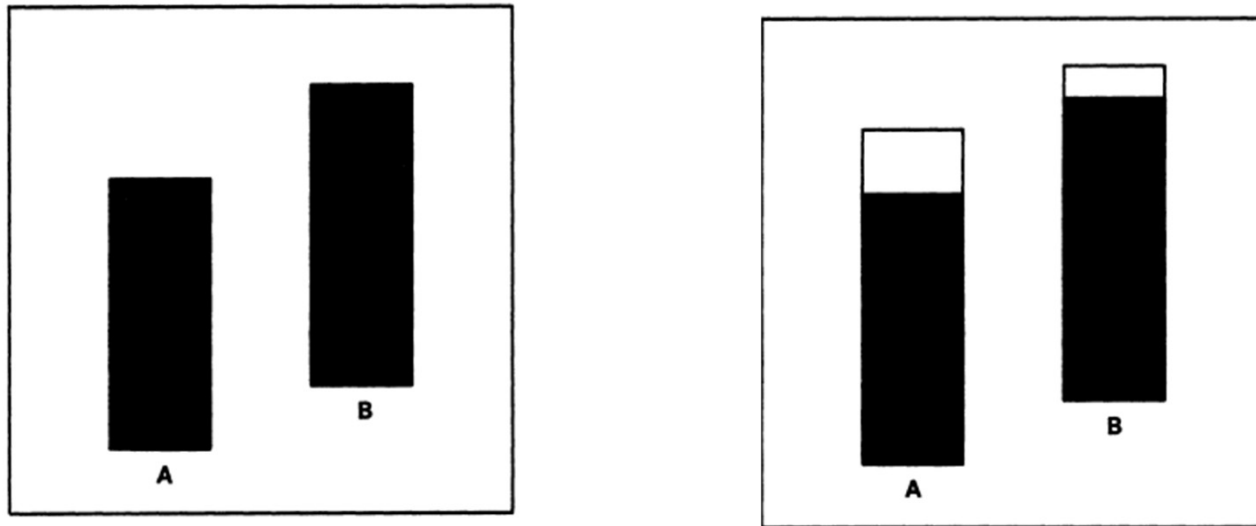
R = Reiz , the German word for stimulus, the standard stimulus.

ΔR = The minimum change in R that could be detected.

k = constant (as seen earlier, k = 1/40 for lifted weights)



Weber's law in visualizations



Which rectangle is longer? (e.g. length encodes Money)

Graphical perception: Theory, experimentation, and application to the development of graphical methods
Cleveland, William S and McGill, 1984

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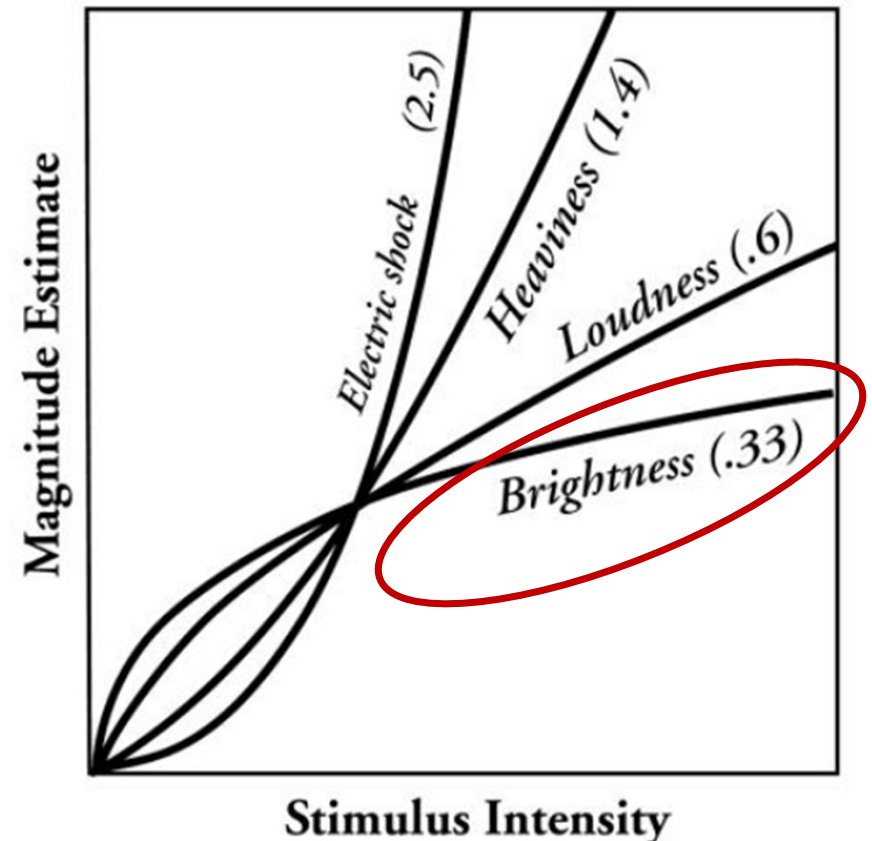
Stevens Law

- Stevens, Stanley Smith (1906-1975) Harvard psychologist
- Stevens proposed that subjects could quantitatively estimate the magnitude of a physical stimulus
- Stevens' magnitude estimation procedure: Provide “modulus” (baseline) stimulus (e.g., loud tone, electric shock) and call it 100. Have subject estimate intensity of subsequent stimuli on scale 0 ... 100, e.g. if tone sounds half as loud rate it 50

Stevens Law

- Stevens, Stanley Smith (1906-1975)
Harvard psychologist
- Stevens' magnitude estimation method yields power relations for many stimuli
- Ψ is the sensation, ϕ is the stimulus, and α is a sense-specific constant

$$\Psi = \phi^{\alpha}$$



Examples of Stevens' Law in Visualizations

1. Bar Charts (Length)

- **Exponent $n \approx 1.0$:** People are generally **good at judging length**. If a bar is twice as long as another, it's typically perceived as twice as much.
-  Good perceptual accuracy.
-  Recommended for representing quantitative data.

2. Bubble Charts / Circle Size (Area)

- **Exponent $n \approx 0.7$:** People **underestimate differences in area**. If one circle is 4x the area of another, it might only be perceived as 2.5x larger.
-  Can lead to misinterpretation of magnitude.
- Use with caution or add numerical labels.

Examples of Stevens' Law in Visualizations

3. 3D Volume-Based Visuals

- **Exponent $n \approx 0.6$** : People are **even worse at judging volume**, so using spheres or cubes to represent data exaggerates or flattens differences.
-  Generally discouraged.

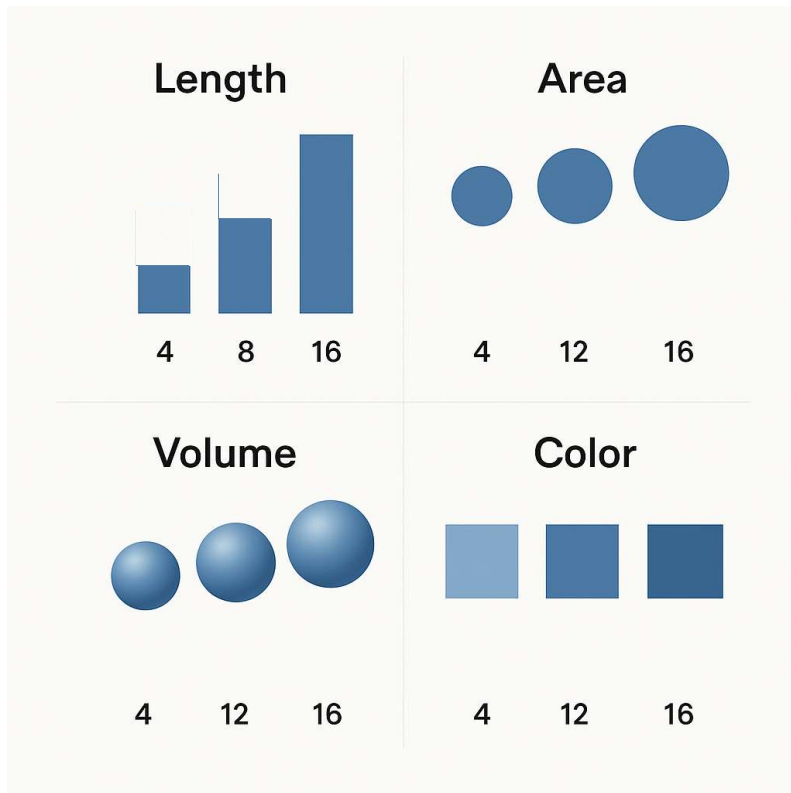
4. Color Intensity / Saturation

- **Exponent $n \approx 0.3-0.5$** : Perception of color brightness or saturation does not scale linearly with the actual value.
- Use **color gradients** carefully, preferably for categorical or ordinal data, not precise values.

5. Position on a Common Scale (e.g., Dot Plots)

- **Exponent $n \approx 1.0$** : Like bar length, **position** is perceived accurately.
-  Excellent choice for comparing values.

Examples of Stevens' Law in Visualizations



Stevens' Law is about **perceived magnitude $\neq$ actual magnitude**. The key point is: The way we **perceive** differences in length, area, volume, and color does **not scale linearly** with the actual data values.

Type	Actual Data	Physical Scale Used	Perceived Value (Approx)
Length	4 $\rightarrow$ 8 $\rightarrow$ 16	Doubles each time	4 $\rightarrow$ 8 $\rightarrow$ 16 (perceived ~accurate)
Area	4 $\rightarrow$ 8 $\rightarrow$ 16	Area doubles	4 $\rightarrow$ 6.5 $\rightarrow$ 10 (underestimated)
Volume	4 $\rightarrow$ 8 $\rightarrow$ 16	Volume doubles	4 $\rightarrow$ 6 $\rightarrow$ 8 (even more underestimated)
Color	4 $\rightarrow$ 8 $\rightarrow$ 16	Brightness doubles	Hard to distinguish (especially pastel tones)

Summary

- Stevens' Law explains how big something feels.
- Weber's Law explains how small a change we can detect.

Mental Models

A mental model is a functional analogue representation to an external interactive visualization system with the following characteristics:

- The structural and behavioral properties of external systems are preserved in mental models.
- A mental model can preserve schematic, semantic or item specific information about the underlying data.
- Given a problem, a mental model of an interactive visualization can be constructed and simulated in working memory for reasoning.

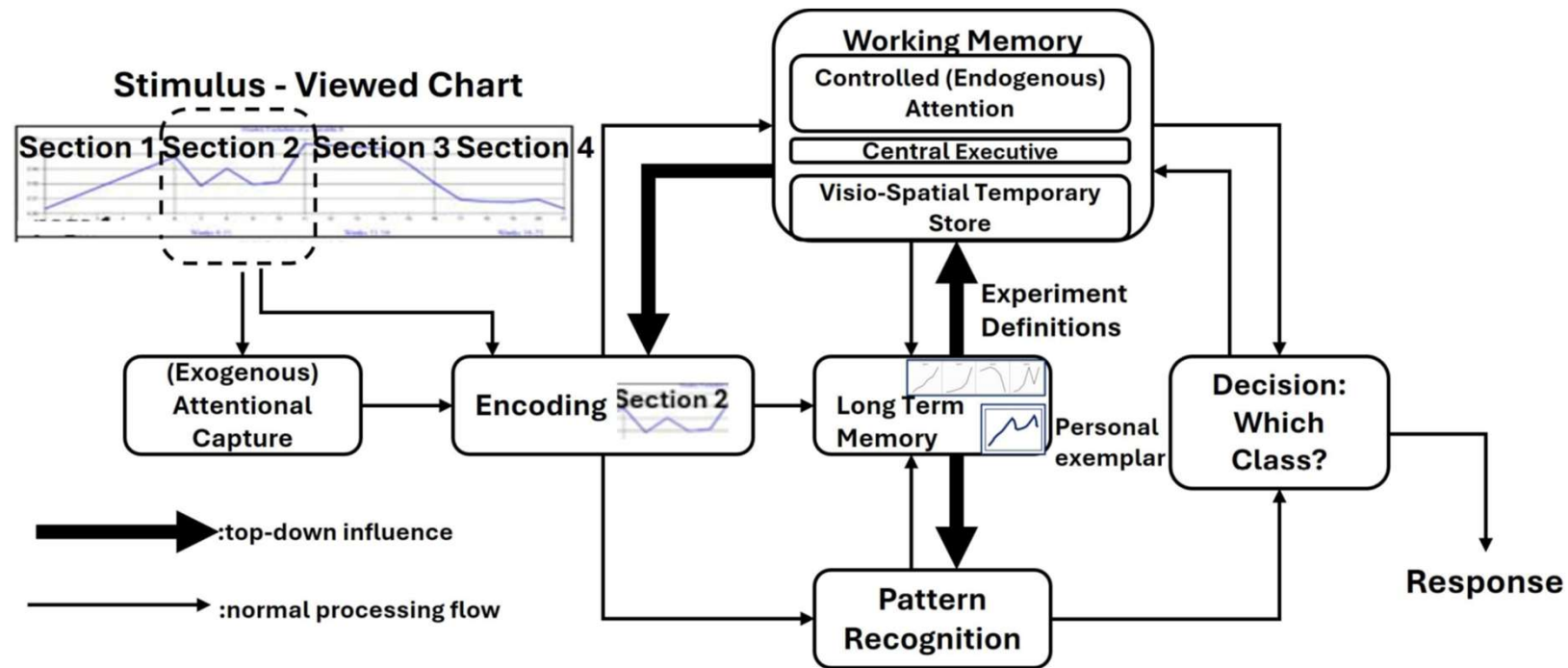
Mental models/Exemplars

- Exemplar theory suggests that category judgments are made by comparing new stimuli to stored exemplars.
- Mental models, influenced by both task demands and individual differences, play a role in visual classification

[1] R. M. Nosofsky. Attention, similarity, and the identification–categorization relationship. *Journal of experimental psychology: General*, 115(1):39, 1986.
[2] R. M. Nosofsky. Exemplar-based accounts of relations between classification, recognition, and typicality. *Journal of Experimental Psychology: learning, memory, and cognition*, 14(4):700, 1988.

Mental models/Exemplars

Cognitive framework [1] for classifying patterns in an experiment [2].



[1] R. E. Patterson, L. M. Blaha, G. G. Grinstein, K. K. Liggett, D. E. Kaveney, K. C. Sheldon, P. R. Havig, and J. A. Moore. A human cognition framework for information visualization. *Computers & Graphics*, 42:42– 58, 2014.

[2] S. Eisler, J. Meyer - The classification of patterns in visualizations: effects of data complexity and individual differences – *IEEE Transactions on Visualization and Computer Graphics* (submitted)

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Mental Models

- External Anchoring: projection of mental forms onto visual forms
 - The user has to know the meaning of the symbols
 - Is the process by which a user projects pre-existing mental structures (knowledge, expectations, symbols) onto a visual form in order to understand it.
 - When looking at a visualization, the viewer "anchors" their internal understanding to the symbols or visuals they see — but only if they understand what the symbols mean.

External Anchoring examples

- If you show a user a **line chart**, most people will *automatically interpret* it as showing **change over time** — because that's a common mental model for how lines work in data viz.
- Show them a **red downward arrow** — they'll often interpret it as **something bad decreasing** (like profit or stock value) — again, a mental association.
- But if the symbols are **unfamiliar** or **not explained** (e.g., triangles, abstract shapes, or even cultural icons), **the user cannot anchor their internal understanding** — and the visualization fails to communicate clearly.

Mental Models

- Information Foraging: strategies and technologies for information seeking, gathering, and consumption adapted to the flux of information in the environment. May depend on curiosity.
 - Restructure the data and visualize to find interesting patterns
 - Exploration

*Inspired by animal foraging theory, Information Foraging Theory (Pirolli & Card) suggests that users seek, gather, and consume informational "prey" — things like patterns, insights, or anomalies — much like animals look for food.

Information Foraging in Visualizations

Key Aspects in Visualization

1. Curiosity-Driven Exploration

- Users *don't always know* what they're looking for.
- They follow “**information scent**” — cues that something interesting or useful lies ahead.
 - Example: seeing a spike in a line chart may invite deeper drilling.
-  **Design implication:** Good visualizations support *open-ended exploration*, not just static answers.

Information Foraging in Visualizations

2. Restructuring Data to Find Patterns

- Users reorganize how data is presented to **reveal new insights**.
 - Sorting
 - Filtering
 - Zooming/panning
 - Grouping by attributes
-  **Design implication:** Tools like dashboards, pivot tables, or interactive charts help users reshape the view and “hunt” more effectively

Information Foraging in Visualizations

3. Information Flux and Environment

- The “environment” is often **overwhelming** (big data, real-time updates).
- Users need help deciding **where to focus their attention**.

Design implication:

- Use pre-attentive cues (e.g., color, position) to guide the eye.
- Provide “preview” mechanisms (hover, mini-maps, summaries).

Visual Exploration as Foraging

Foraging Action	Visualization Equivalent
Scanning the field	Glancing over a dashboard
Sniffing for scent	Spotting a visual anomaly (e.g., outlier)
Moving closer	Zooming into a time range or region
Extracting nutrients	Interpreting or exporting an insight

Mental Models

- Cognitive Offload: Create new model through insights (symbolic model)
- Visualizations help users offload complex mental processes onto the visual display, freeing up working memory and enabling them to form new, internalized mental models based on insights they discover.

Cognitive Offload:

Cognitive offloading is when we **use external tools** (e.g. notes, diagrams, maps, or data visualizations) to help us:

- Remember things
- Compare options
- See relationships
- Detect patterns

In visualization, the chart becomes an **external extension of the mind**. Instead of calculating or remembering everything, users “offload” those burdens onto the visualization.

Cognitive Offload:

When users explore a well-designed visualization:

1. They **recognize patterns** (like correlations, trends, outliers)
 2. These patterns **inform new internal models** of how the system works (e.g. “sales rise after marketing spend increases”)
 3. These new models are often **symbolic or conceptual** — users can now talk or reason about them
- This is the “**create a new model through insights**” part — the user internalizes **symbolic understanding** of the system behind the data.

Experiment – short assignment

- 15 to 20 minutes experiment
- Related to mental models in visualizations
- Send me via email a print screen of each one of the 5 plots you drew

Perceptual Processing

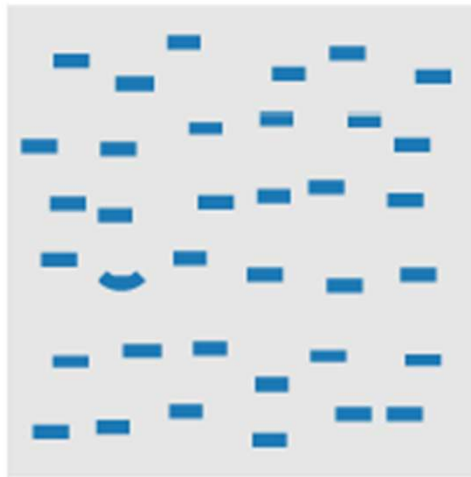
Perception can be:

- **intrinsic and uncontrolled (pre-attentive)** fast and parallel, often within 250 ms
- **controlled (attentive)** slower, uses short-term memory, selective and can represent aggregates of what is in the scene

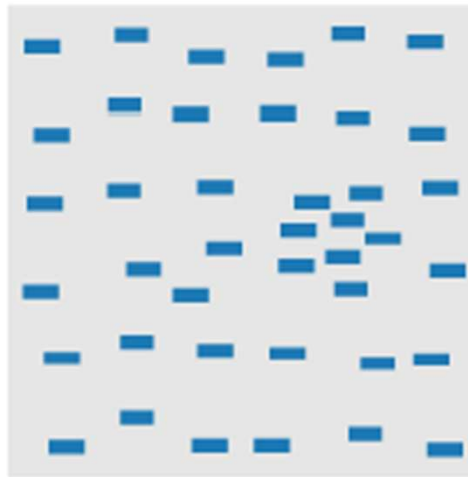
Pre-attentive processing

- Why is this important? In displaying information, it is often useful to be able to show things “at a glance.”
- If you want people to be able to identify instantaneously some mark on a map as being of type A, it should be differentiated from all other marks in a pre-attentive way.

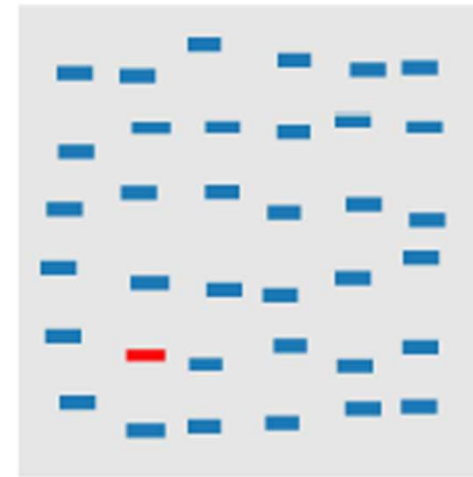
Pre-attentive - Perceptual Processing



Curvature



Density



Hue

Healey, C. G., & Enns, J. T. (2012). Attention and Visual Memory in Visualization and Computer Graphics .
IEEE Transactions on Visualization and Computer Graphics, 18 (7), 1170-1188

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Pre-attentive - Perceptual Processing

- Preattentive processing asks in part, “What visual properties draw our eyes, and therefore our focus of attention to a particular object in a scene?”
- An equally interesting question is, “What do we remember about an object or a scene when we stop attending to it and look at something else?”
- Evidence suggests that a viewer’s current state of mind can play a critical role in determining what is being seen at any given moment, what is not being seen, and what will be seen next.

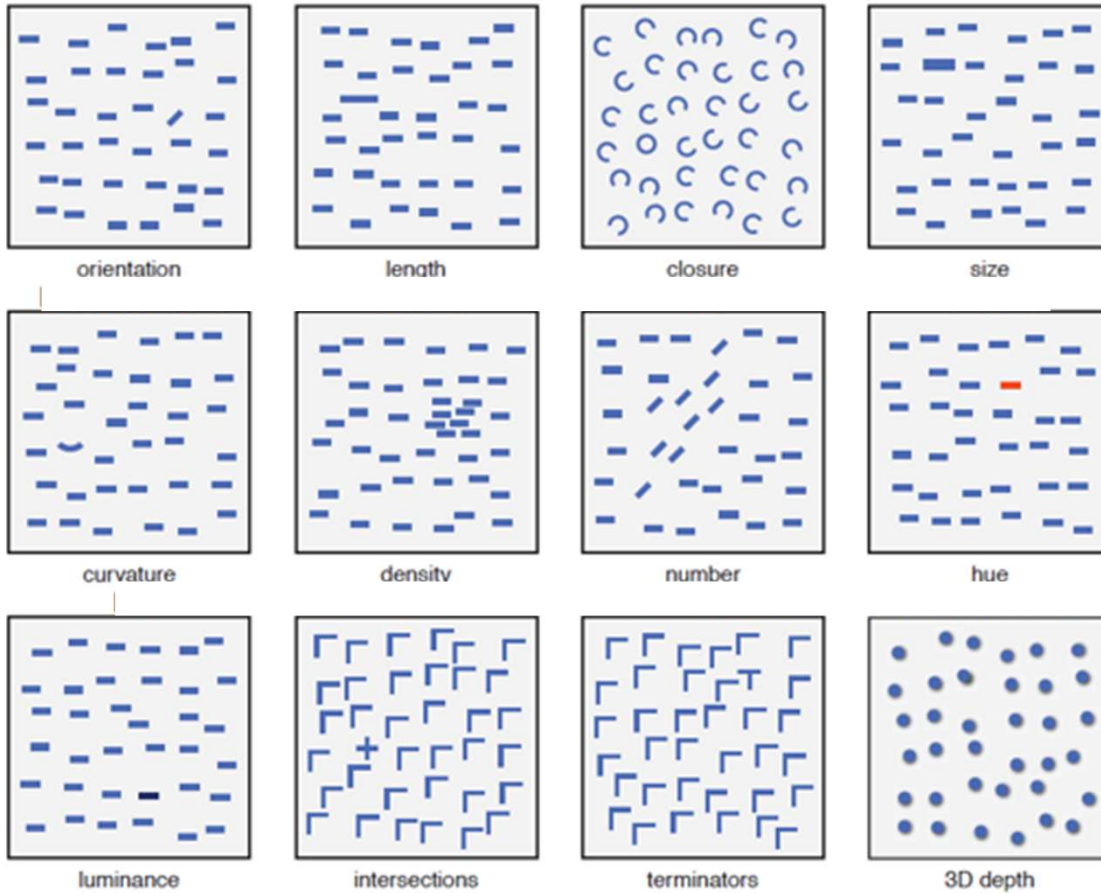
Pre-attentive Processing

Visual Features

- Length
- Width
- Size
- Curvature
- Number
- Terminators
- Intersection
- Closure
- Color

Visual Tasks

- Target detection - detect the presence or absence
- Boundary detection - detect a texture boundary between two groups
- Region tracking - track one or more elements with a unique visual feature as they move
- Counting and estimation - count or estimate the number of elements with a unique visual feature



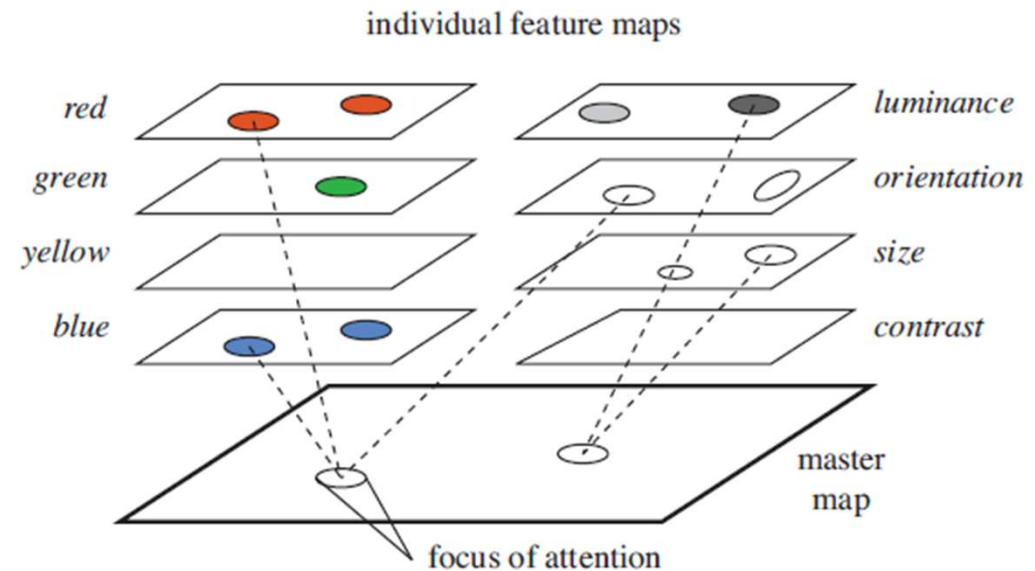
Healey, C. G., & Enns, J. T. (2012). Attention and Visual Memory in Visualization and Computer Graphics .
 IEEE Transactions on Visualization and Computer Graphics, 18 7), 1170 1188

Theories of Pre-attentive Processing

- Feature integration theory (FIT)
- Texton theory
- Similarity theory
- Guided search theory
- Boolean Map Theory
- Ensemble Coding
- Feature Hierarchy

Feature integration theory (FIT)

- Treisman's feature integration model of **early vision**—individual maps can be accessed in parallel to detect feature activity, but focused attention is required to combine features at a common spatial location



Feature integration theory (FIT)

- When the human visual system first sees an image, all the features are encoded in parallel into their respective maps. A viewer can access a particular map to check for activity, and perhaps to determine the amount of activity.
- However, the individual feature maps give no information about location, spatial arrangement, or relationships to activity in other maps.
- This framework provides a general hypothesis that explains how pre-attentive processing occurs. If the target has a unique feature, one can simply access the given feature map to see if any activity is occurring. Feature maps are encoded in parallel, so feature detection is almost instantaneous.

Feature integration theory (FIT)

- First, Treisman tried to determine which visual properties are detected pre-attentively. She called these properties “pre-attentive features”.
- Second, she formulated a hypothesis about how the human visual system performs pre-attentive processing. This means that she did an experiment...
- Measured pre-attentive task performance in two different ways: by response time and by accuracy.

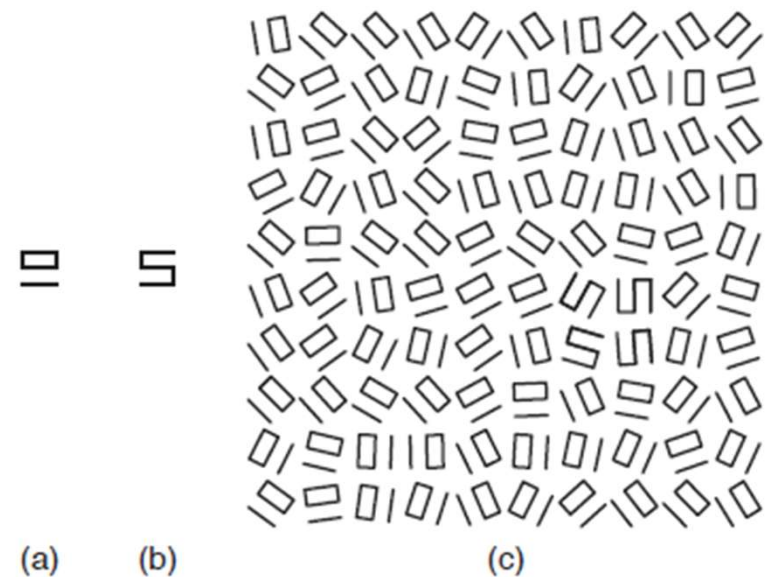
Feature integration theory (FIT) – experiment for response time model

- Viewers are asked to complete a target detection task as quickly as possible and maintain accuracy.
- The number of distractors is repeatedly increased. Then if the completion time is relatively constant and below some chosen threshold, independent of the number of distractors, the task is said to be pre-attentive.
- Hypothesis: If the task is not pre-attentive, viewers need to search serially through each display to confirm a target's presence or absence. ➔ Increasing the number of elements in the display will produce a corresponding increase in the completion time.

Textons – Bela Julesz

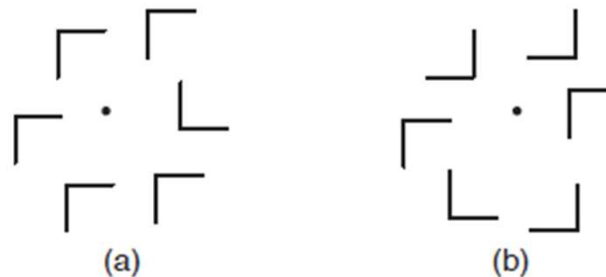
Early visual system detects three categories of features called textons:

1. Elongated blobs—lines, rectangles, or ellipses— with specific hues, orientations, widths, and so on.
2. Terminators—ends of line segments.
3. Crossings of line segments.



Similarity Theory - Duncan and Humphreys

- Hypothesized that search ability varies continuously and depends on both the type of task and the display conditions.
- Search time is based on two criteria: T-N similarity and N-N similarity.
 - T-N similarity is the amount of similarity between targets and nontargets.
 - N-N similarity is the amount of similarity within the nontargets themselves.



N-N similarity affecting search efficiency for an L-shaped target:

(a) high N-N (nontarget-nontarget) similarity allows easy detection of the target L;

(b) low N-N similarity increases the difficulty of detecting the target L

Guided Search Theory - Jeremy Wolfe

- Incorporates the goals of the viewer into a model of human search
- An activation map based on both bottom-up and top-down information is constructed during visual search
- Early vision divides an image into individual feature maps

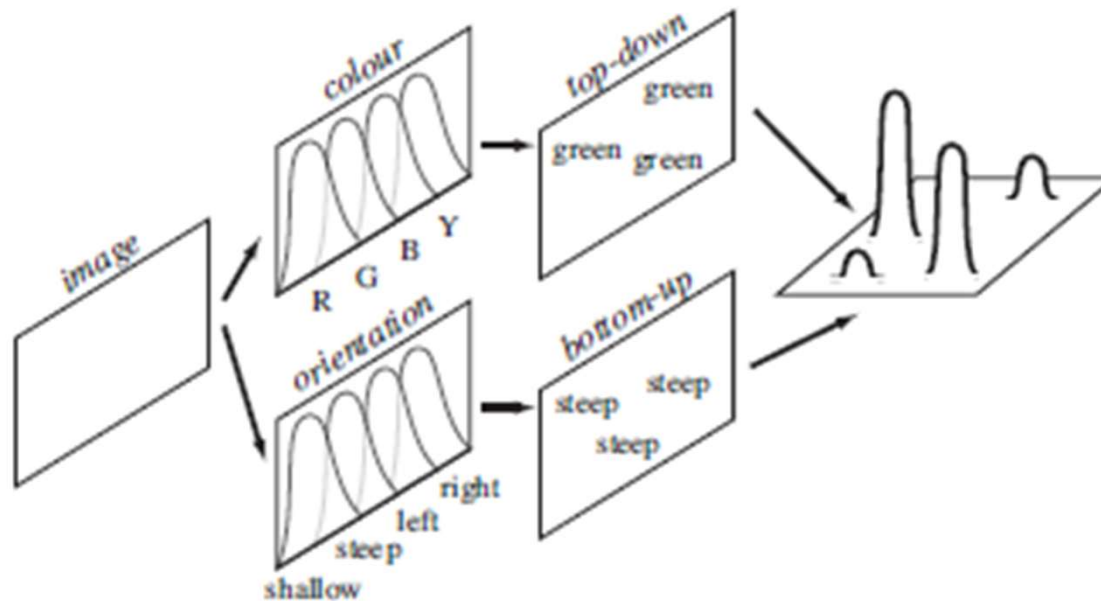
Guided Search Theory - activation map

- An activation map refers to a visual representation of the parts of an image that are most important for a particular task, such as detecting edges or recognizing faces.
- Activation maps are generated by analyzing the responses of neurons in the visual cortex of the brain to specific visual stimuli. These neurons are highly specialized and respond to specific features in the visual scene, such as edges, corners, or colors.

Bottom-up and top-down processing

- The interaction between bottom-up and top-down processing is crucial for human perception and cognition.
- Bottom-up processing provides the raw sensory information needed to perceive the environment.
- Top-down processing helps to fill in missing information, identify patterns, and make sense of the sensory input in the context of prior knowledge and expectations.

Guided Search Theory - Jeremy Wolfe



Guided search for steep green targets, an image is filtered into categories for each feature map, bottom-up and top-down activation “mark” target regions, and an activation map combines the information to draw attention to the highest “hills” in the map

Perceptual Salience

- How should I represent the data?
- How can I highlight important data values when they appear?
- How can I ensure that viewers perceive differences in the data accurately?
- Having a given chart – what is the most salience feature?

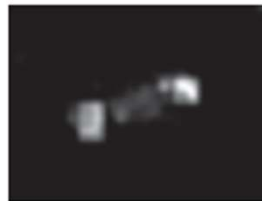
Original Image



Fixation Density Map

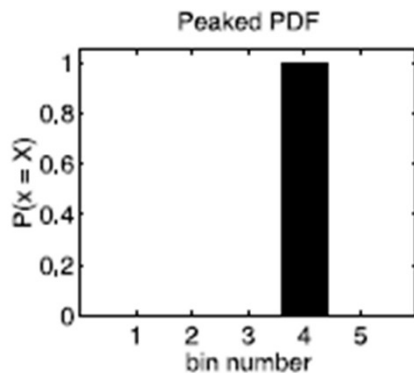


Using local entropy

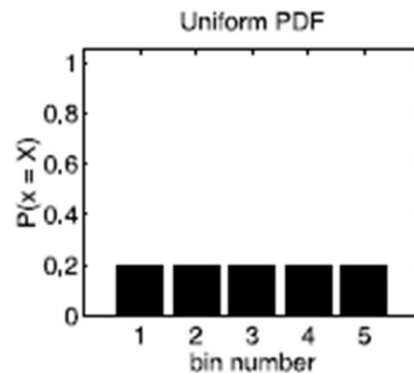


A Computational Model for Saliency Maps by Using Local Entropy
Yuewei Lin - 2010

Quantifying Visual Information



(a) MINIMUM ENTROPY
PREDICTABLE
STRUCTURED



(b) MAXIMUM ENTROPY
UNPREDICTABLE
COMPLEX

To calculate entropy for a discrete function, given by:

$$H = - \sum_{i=1}^N p_i \log_2(p_i)$$

where i is the bin number, N is the number of populated bins, and p_i is the proportion of data falling into the i th populated bin, subject to the condition: $\sum_i^N p_i = 1$.

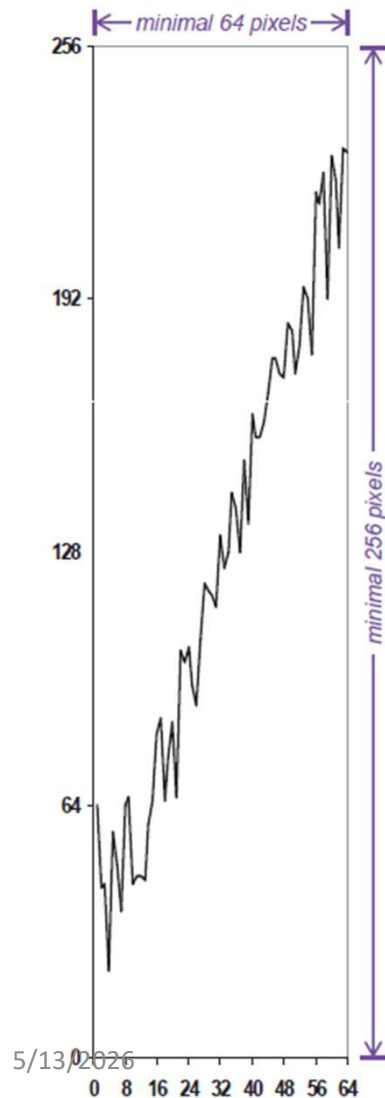
- Minimum entropy corresponds to a sharply peaked probability distribution, such as that shown in a. ($H_{min} = -P_4 \log_2(P_4) = 0$.)
- Maximum entropy corresponds to a uniform probability distribution, shown in b. Every bin is populated with an equal number of data points, such that $p_i = \frac{1}{N_b}$ for all i , where N_b is the total number of bins.

$$H_{max} = \log_2(N_b)$$

- Relative entropy:

$$H_R = \frac{H}{H_{max}}$$

Quantifving Visual Information



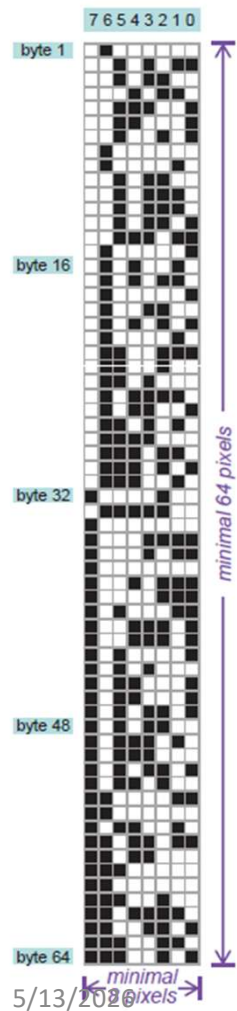
- 64 independent samples in the time series
- Each sample has an integer value range between 0 and 255 (2^8)
- Samples are taken at a regular temporal step. The visualization displays 64 pixels corresponding to the random samples, and the connecting lines between consecutive samples.
- The probability mass function of each sampling value is independent and identically-distributed : $p = 1/2^8$

$$H = - \sum_{i=1}^{64} \sum_{j=0}^{255} \frac{1}{2^8} \log_2 \frac{1}{2^8} = 512$$

- Expected as the time series requires a minimum of 64 bytes (i.e., 512 bits) to encode.

Source: An Information-theoretic Framework for Visualization, Min Chen (2010)

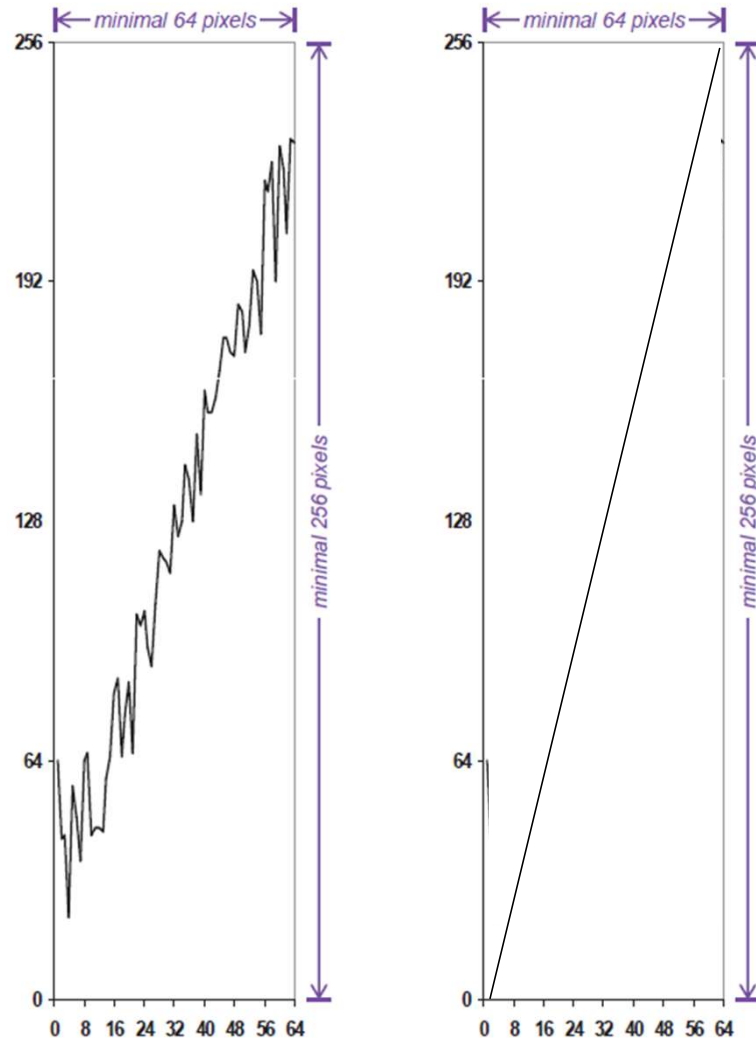
Quantifying Visual Information



- In theory, we can also display this binary code as shown.
- This would require a huge perceptual and cognitive load to decode if not impossible.
- This uses more than 1 pixel for each “bit” box, as the figure would otherwise be unreadable by human eyes.
- If only 4×4 pixels per “bit” box, it would result in a total of 2^{13} bits = $16 \times 8 \times 64 = 8192$.
- In practice better to use the previous plotting area with 256×64 black-white pixels (2^{14} bits) as minimal requirement.

Source: An Information-theoretic Framework for Visualization, Min Chen (2010)

Quantifying Visual Information

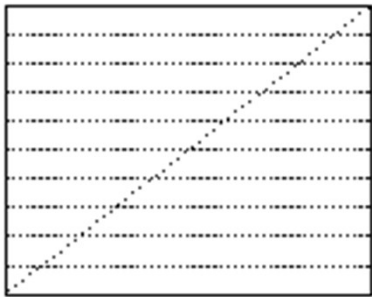


- What is the problem here?
- According to what we saw, both plots would have the same entropy!
- What do we do?

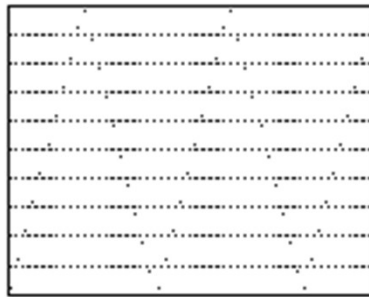
Source: Mays (2002) Information entropy to measure temporal and spatial complexity of unsaturated flow in heterogeneous media

5/13/2020

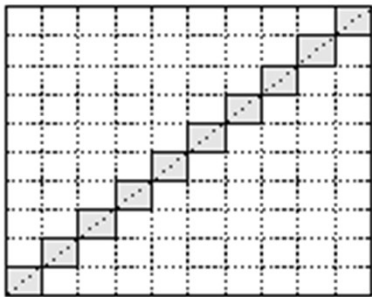
Quantifying Visual Information



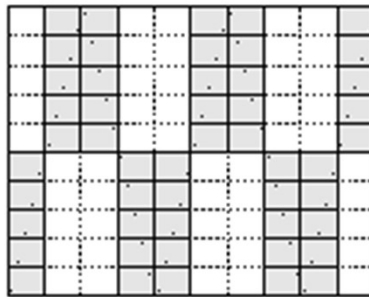
(a)



(b)



(c)

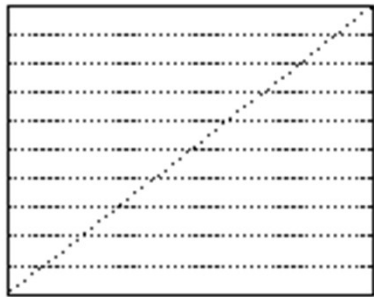


(d)

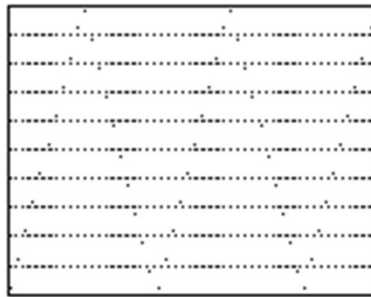
Use multiple dimensions to calculate entropy.
Entropy calculation in multiple bins.

- (a) and (b): the data is divided the data is divided along the y axis into rows, making a 1D division. For either linear or oscillating data, the measurements are uniformly distributed among 10 bins, producing a relative entropy of 100%.
- (c) and (d): the data is divided along both the x and y axes, making a 2D bin division by dividing both the x and y axes into 10 bins each.
- (c): $H = -\sum p * \log(p) = -10 * \frac{1}{10} * \log(\frac{1}{10})$
- (d): $H = -\sum p * \log(p) = -50 * \frac{1}{50} * \log(\frac{1}{50})$
- $H = -\sum p * \log(p) = -20 * \frac{1}{20} * \log(\frac{1}{20})$

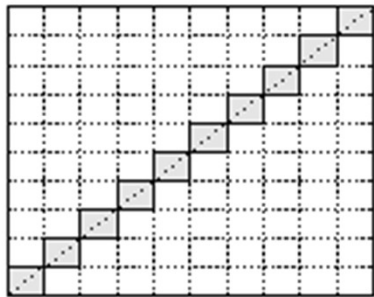
Quantifying Visual Information



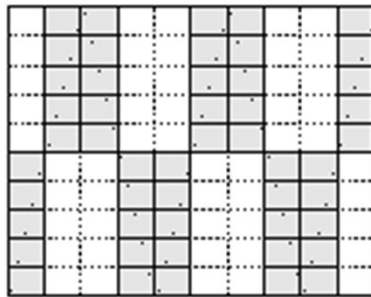
(a)



(b)



(c)



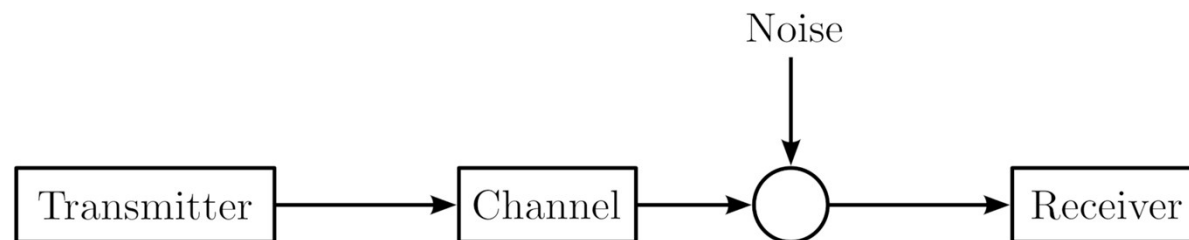
(d)

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Resource Model of Human Information Processing

- Channel capacity from the field of information theory (George Miller, 1956)
- Miller measured it in bits (borrowing again from information theory), depending on the number of levels that the average human could measure with high accuracy. Thus, if errors routinely begin when more than 8 levels of a phenomenon are tested, the channel capacity for this phenomenon is 3 bits.



Papers to read

- Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods, William S. Cleveland; Robert McGill – 1984
- Mental Models, Visual Reasoning and Interaction in Information Visualization: A Top-down Perspective, Zhicheng Liu and John T. Stasko – 2010
- A Computational Model for Saliency Maps by Using Local Entropy
Yuewei Lin - 2010